Augmented Reality for Experience-centered Spatial Design

A quantitative assessment method for architectural space

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While interaction and product design are using data-driven methods on a daily basis, not only similar data but also methods for making use of such data for designing architectural space is missing. The spatial design feedback loop is broken, as we lack quantitative assessment systems that reveal experience and interaction from the user's perspective. This paper describes the first of its kind development of a spatial assessment and design methodology utilizing Augmented Reality headsets, aiming to fill the gaps in the spatial design iteration loop.

Keywords: *Augmented Reality, Behaviour Tracking, Feedback Loop, Occupancy Observation, Mixed Reality*

INTRODUCTION

When designing, architects use their tacit knowledge with intuitions accumulated over years. However, compared to other design fields, such as interaction design, there is not much measurable feedback to tell architects if their design is well perceived by users.

Other design disciplines use a more defined iterative design process – for example, a product goes through loops of Design - Prototype - Assess (Gossain and Anderson, 1990). User's interaction data with prototypes are collected and analyzed for design optimization. However, this iteration loop is broken in spatial design (Figure 1). The optimization of spatial design normally goes through phases of Design - Visualize - Assess. Compared to product or UX design (Thompson et al., 2020), the user's feedback in spatial design is often ambiguous, retrospective or limited to verbal expressions, due to the lack of an empirical measuring and assessing method, and also the lack of a spatial feeling and presence in 2D visualization (Hermund et al., 2017).

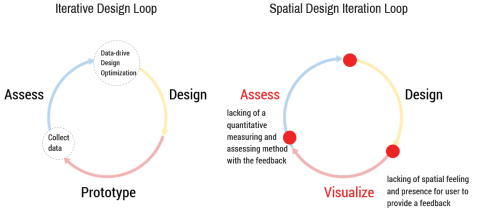


Figure 1
Left: Iterative Design Feedback Loop. Right: Spatial Design Feedback Loop.

The current state of the art research investigates the use of Virtual Reality headsets in simulating the entire architectural space and observing users' responses through tracking of eye-gaze, movement of people and EEG monitoring (Shemesh et al., 2017), in order to improve the space. We find that VR is

however of limited use in the assessment of architectural design, as the required user's immersion is limited by the quality of the synthetic 3D environment and the user's inability to experience space in bodily movement (Heydarian et al. 2015). Augmented Reality headsets on the other hand are by nature immersive (Carmigniani and Furht 2011). We find that these have the potential to collect more intuitive reactions of the users towards the proposed design, as it modifies the world that users are physically moving in. Understanding that a majority of architecture projects takes place in existing urban spaces or the interior, AR bears the potential for novel design-related dialogue and data collection.

This paper presents the development of a spatial assessment and design methodology utilizing the embedded tracking functions of Augmented Reality headsets to collect occupants' feedback. Combined with technologies in space scanning, point cloud processing and data visualization, we aim to develop and test a workflow to fill the gaps in the feedback loop, enabling architects to engage users and speculate occupancy in an early stage of design.

FRAMEWORK

The proposed method is developed to facilitate spatial design optimization, ideally in the native software environment for designers in order to allow for ease of use. Draft design is to be placed in space as holograms, and users will be asked to navigate and complete certain tasks in the augmented environment. Behaviour data including circulation, attention and view recording are to be collected and visualized to help designers identifying the problems in the navigation experience and verifying the efficacy of design decisions. Those outcomes can then inform the next design iteration(Figure 2). The following experiments describe the motivation and development of the framework and the observations in terms of design feedback.

Workflow

Hardware: Faro scanner, Microsoft Hololens2

Software: Faro *Scene*, Rhino, grasshopper(plugin: *Fologram*, *Volvox*)

The physical environment is to be scanned and represented as point clouds, which will be the base for behaviour mapping but will not be visible in AR.

To align the point clouds to HoloLens's internal spatial mapping, the site is to be scanned by both Faro scanner and Hololens2. Scan result from

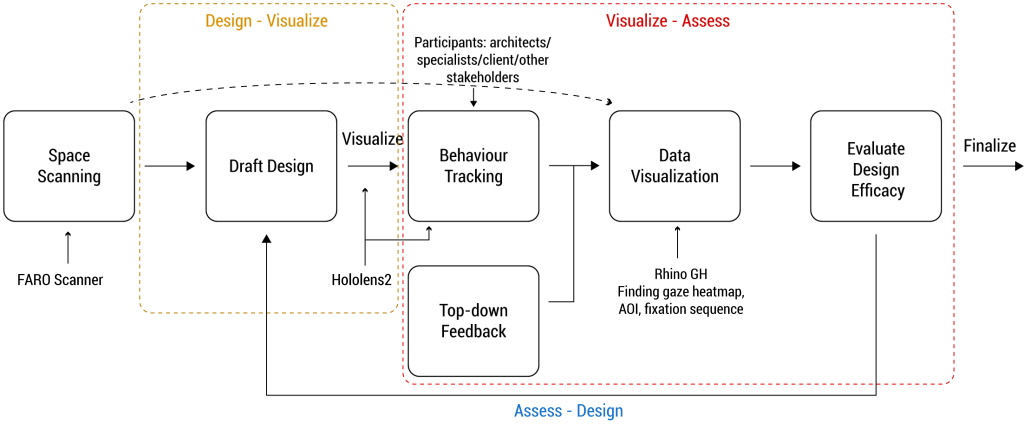


Figure 2
Refined feedback loop workflow

Hololens2 is exported as mesh in Rhino with Fologram Grasshopper plugin. Scan results from FARO are registered and colourized with FARO Scene and imported into Rhino as point clouds, aligned to the mesh scan from Hololens2 as the base for tracking visualization.

The design intervention is modelled in 3D software and visualized in Hololens2 through Fologram Plugin in Rhino. The test subjects are to be equipped with Hololens2 and to navigate the space freely or with certain tasks. Hololens2 is connected to Rhino with the *Fologram* plugin in grasshopper so participants' positions and gaze directions were recorded and stored. The tracking marks the headset's position points and orientation vectors every 20ms. The position points are visualized with small spheres of gradient colours to indicate the sequence of movement.

A more detailed understanding of users attention to the space around them could be achieved with eye-gaze tracking. This is supported by Hololens2, however, that data can only be extracted through software (eg. Unity with MTRK) (Microsoft 2019) outside the design environment. However, the following studies show that the orientation vector from tracking devices in Fologram, with appropriate calibrations, can indicate the gaze direction accurately enough for application on an architectural scale. Therefore we find it is justified to minimize the complexity of the workflow and use a calibrated orientation vector only.

For lighter data processing, environment point clouds are voxelized with the *Volvox* plugin in Grasshopper, and the first intersection of the gaze vectors with the voxel mesh or with the hologram breps are found to indicate where gaze falls. Based on the distance between intersection points and mesh vertices, We then developed a heat mapping method that visualizes where users' attention falls, with coloured dots evenly distributed as an overlay on the environment model and virtual objects. In this way, both the stimulus and the level of attention can be clearly seen. Depending on the task, gaze information is also visualized into areas of interest and fixation sequence, which will be detailed in the Museum study.

METHOD & EVALUATION CRITERIA

Three studies were conducted to test the framework against design use cases on a technological level. The development of the framework and toolset has taken place with an agile method, meaning that the findings of each iteration inform the next use case. These use cases have been chosen by an increasing degree of complexity for both the AR technology, as well as the developed framework:

- Interior design - small and large scale
- Outdoor space
- Renovation project

Parallel to the tracking data a video of the users movements in space was recorded within the AR device. This served as ground truth for the evaluation of the tracking quality.

STUDY 1 - INTERIOR DESIGN

Study 1 aims to test the scanning capacities of different scanning devices and developing a method of converting tracking data to clear visual feedback. To test spaces of different scales, a 25sqm classroom and a 1240sqm auditorium were chosen as sites. Holograms of varying sizes were placed on different heights to test the spatial mapping and visualization capacity of Hololens2.

Outcome and reflection

The scanning distance of Hololens2 is limited to around 4.5m, therefore additional scanning devices are required to obtain the environment model. For the experiments in this paper, a FARO scanner is used to obtain point clouds of the existing environment. Holograms of varying sizes and locations are displayed correctly, even when the distance to the user is further than the scanning limit.

The coloured dots are overlaid on the scanning result of the physical environment (Figure 3) and virtual objects (Figure 4) which clearly shows where



Figure 3
Gaze Heatmap overlaid on (voxelized) environment scan

the gaze stayed longer and where details are overlooked. The result corresponds to the video recorded in Hololens2, so this method can be a valid indication of attention-tracking.

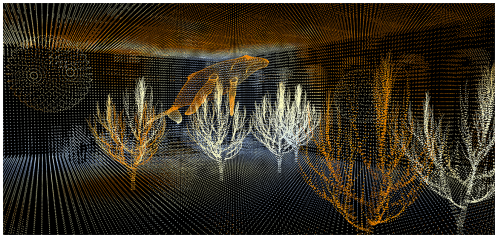


Figure 4
Gaze heatmap on both (voxelized) environment scan and virtual objects (base model hidden).

STUDY 2 - OUTDOOR SPACE

Study 2 focuses on observing the impacts of virtual interventions on people's spatial behaviour and testing the workflow for evaluating design efficacy. The site was chosen on an open square, and a series of virtual walls were set up as a draft exhibition design example. The virtual walls were designed with a proposed visiting circulation path and modelled with timber texture. With data recorded in Hololens2, the case study observes if the designed walls can direct people to follow the proposed path and find the exit without signs, and where are the problems in navigation.

Outcome and reflection

The virtual wall setup with timber material and shadow was displayed in AR (Figure 5 left). The hologram walls show a strong sense of presence as people are following the interventions instead of walking through the holograms. The heat map from recorded data is overlaid on the virtual wall, and in this study, was made into animation in the movement sequence. The pattern of space perceiving can be understood from the heatmap and animation, as the movement speed and time of focusing is shown with the animation. However, in the analysis of design efficacy, it is unclear about the cause of the gazing. The questionnaire can be cross-referenced to give indications on the causes.

With the method described in chapter2.1, the heatmap visualizes the visual attention on the virtual walls (Figure 5 right) and series of gradient-coloured dots indicates visiting paths. It is clear that areas at the intersections receive a lot of attention and witness back and forth in moving. This method of collecting and visualizing circulation and gaze data provides a quick overview of visitors' navigation experience and helps to identify areas that attract more attention. However, for a more specific design efficacy analysis, the cause of the gazing is unclear. A retrospective questionnaire can be employed for cross-referencing where qualitative information is needed.

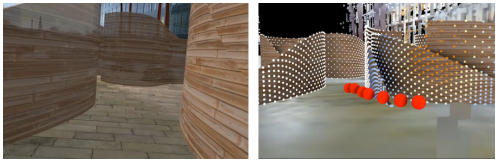


Figure 5
Left: view recorded from Hololens2.
Right: Gaze heatmap overlay

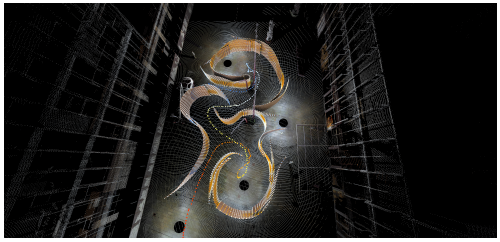


Figure 6
Heatmap on Exterior Hologram setup

STUDY 3 - MUSEUM STUDY

The study takes place in the Ny Carlsberg Glyptotek for the ongoing renovation of this listed exhibition space in Copenhagen, Denmark. In this study, the current gallery setup was modified by the virtual hologram setups: a proposed juxtaposition of modern and classic Danish Art was designed, by repositioning some existing artworks and adding contemporary artworks in an augmented environment (Figure 7).

Table 1
Participates profile

Participate	Age	Gender	Education Background	Experience with AR
A	18	Male	Highschool	No
B	22	Male	BA - Music	No
C	25	Female	MA - Strategic Design	No
D	24	Female	MA - Architecture	Yes

Visitor's behaviour data under both existing environment and augmented setups were tracked, visualized and analyzed for further design reference using methods described in Chapter 2.1. Tests were conducted under COVID restriction by 4 participants, with different academic backgrounds, ageing from 18 to 25 (Table 1). Without instructions on time limits, participants were asked to visit the gallery freely for as much time as they would like.

Additionally, since most AR visualizations are adding elements to space, we also attempted to visually subtract objects by using holograms to cover existing objects (Figure 8). In Figure 8 (left), by overlaying the floor and wall material, the red separation wall appears to be transparent. In Figure 8 (right), the mint walls and paintings on top are holograms, which shows effective covering on the red existing walls and paintings below. This demonstrates expanded possibilities of modifying the built environment with augmented reality. Interactive feedback methods were also tested: scans of the painting are extracted and duplicated so that the user can drag the hologram of the paintings to their preferred positions. Also, the hand tracking data is used to generate pipes when the user is using the gesture of drawing in air, which allows on the spot mark-ups.

Outcomes and reflections

The recorded positions and directions are processed into heatmap with the method described in the framework. In addition, as a widely used parameter in eye-tracking analysis (Massaro et al. 2012), Area of interest(AOI) was found for areas where more than 5 points are within distance of 30cm, which indicates that the gaze stays for more than 1s. The fixation sequence is then labelled to show what is prioritized by a participant when they perceive the space (Krajbich et al. 2011).

Table 2
information derived from tracking data

Part	Test 1:Time spent in the room (min)	Number of Areas of Interest	Max Fixation Time(min)	Test 2:Time spent in the room (min)	Number of Areas of Interest	Max Fixation Time(min)
A	02:00	15	0:05	01:10	4	0:10
B	07:58	16	0:20	06:32	n/a	n/a
C	09:30	23	0:31	05:10	10	0:15
D	03:20	14	0:10	02:04	6	0:12

We first compared the navigation pattern of each test subject in the existing environment. We can see that despite that almost everyone roughly follows a clockwise sequence, there are huge differences in the preferred pieces and visiting time. We could not draw a conclusion of a common visiting pattern with this small sample size, but we could observe each individual's perception pattern and roughly identify their interest points. Based on that, we can further extract information about the numbers of stimuli (Area of interest), fixation time on each stimulus, areas that receive the maximum fixation time, etc (Table 2). Those



Figure 7
Existing exhibition (left) Augmented exhibition (right)



Figure 8
'Removing' the separation wall in AR (left). Covering existing walls and displace paintings in AR (right).

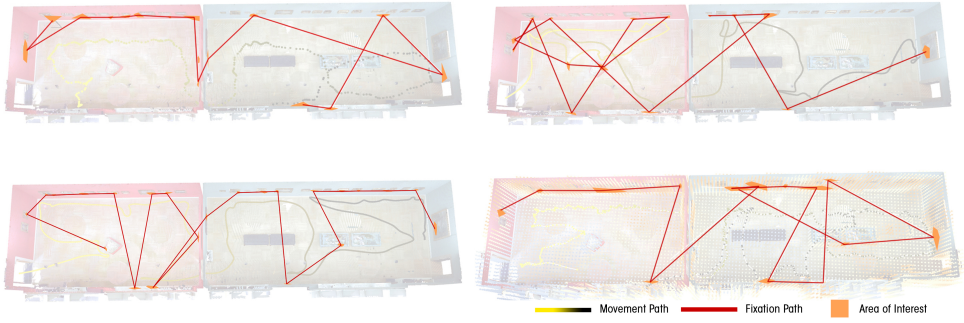


Figure 9
Participate A (left up) : Participate BRight up Left bottom: Participate C Right bottom: Participate D)

data, if collected on a larger scale, can be used to

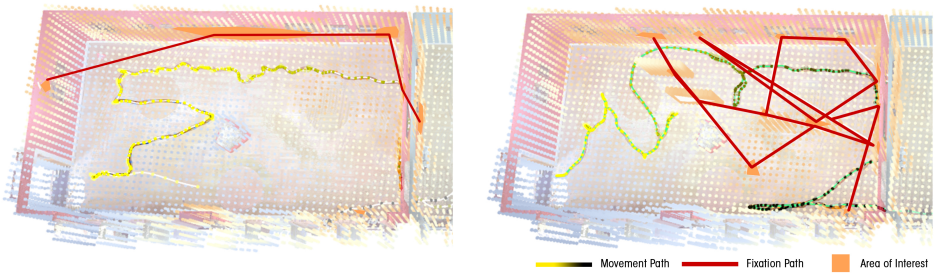


Figure 10
Left: Movement, gaze heatmap, AOI and fixation sequence with existing gallery room. Right: Movement, gaze heatmap, AOI and fixation sequence with augmented gallery room.

speculate visitors' preferences and visiting habits

When comparing the visualized feedback of augmented and non-augmented exhibition spaces, we find significant changes in circulation and focus patterns. Figure10 shows the behaviour data of particpate D before and after adding digital setups. The digital setup seems to attract more attention and the paintings close to the entrance were overlooked and the attention goes to the contemporary artworks first.

However, because of the novelty and interactive nature of AR and hologram, the comparison cannot be valid evidence for evaluating two design set, but more as an observation on how people react to hybrid exhibitions. We could reduce the bias in future experiments by comparing designs that are both displayed as holograms. The other participants show varying levels of willingness for interacting with holograms: Participate B went direct to move and rearrange the holograms, while Participate A and C were more conservative in displacing the setups. We realized our method is limited in recording close interactions like these, which could be expanded in further investigations.

CONCLUSION

The methodology and workflow of spatial design assessment presented in this study show that AR devices can provide a foundation for spatial behaviour studies and a refined design feedback loop. It expands the use case of AR headset to a combination of design, visualization and assessment tool. The movement, areas of interest and fixation sequence has the potential to provide insights to design problems that are overlooked in retrospective evaluations, and therefore architects can better engage users who are not professionally trained in the design phase. This method best fits in experience-centred or site-specific design such as exhibition design, wayfinding design or urban renovation design.

However, the study is limited to a small sample number. Bearing that people's spatial behaviour is heavily influenced by collective behaviours and social interactions(Yehuda, 2012), we are looking into possibilities of studies with larger sample size and social interactions. A limiting factor for the approach is the availability of AR technology. This is currently geared towards individual experiences, as in design meetings. However AR technology might become more common place, as in a speculative setting of a future exhibition, where any user would wear an AR device for the purpose of displaying additional information. This collection of user data would enable a dynamic analysis of users' interest pattern and a coupled tailored visiting experience for each individual. If the method is combined with machine-learning recognition, it has the potential to recognize the specific occupancy interaction behaviour and enable architectural practices to inform their designs

on specific occupant centric parameters (Jørgensen et al. 2020). For larger behavioural studies tracking technologies, such as beacon-based or camera-based tracking, seem more suitable(Jørgensen et al. 2020), though one would not get specific data from a user's gazing.

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